

SARTHAK SRIVASTAVA

Python Backend Developer · Systems Architect · CTO @ DevQBX

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PROFESSIONAL SUMMARY

3rd-year BCA student at VSICS Kanpur (CSJMU) with proven experience architecting high-concurrency asynchronous backend services, event-driven pipelines, and relational database schemas. Hands-on proficiency in Python (AsyncIO, FastAPI, aiohttp), PostgreSQL, SQLite WAL concurrency, and Linux VPS hardening.

TECHNICAL SKILLS

Languages: Python (AsyncIO, Pydantic, SQLAlchemy), SQL, TypeScript, JavaScript, C, Java
Frameworks & APIs: FastAPI, aiohttp, REST APIs, WebSockets, Next.js 16, Unicorn, OpenAPI/Swagger
Databases & Cloud: PostgreSQL, SQLite (WAL mode, busy-timeout queues), Supabase, Redis, Linux VPS Hardening
Systems & Tools: Docker, Git/GitHub, Systemd, APScheduler, Tesseract OCR, PyMuPDF, Render, Nginx

WORK EXPERIENCE

Chief Technology Officer (CTO) · DevQBX · devqbx.in 07/2026 – Present | Kanpur / Remote
• Directing system architecture for 500+ builder community — engineering standards, hackathon platform (QBX Arena), and API integrations.
• Mentoring developers on async backend patterns, API contract design, and production deployment reliability.

Project Manager & Systems Architect · Sitekraft.dev · sitekraft.dev 03/2026 – Present | Remote
• Architected BulkBeat TV Telegram engine; engineered 22-rule AI scoring and <5s alert latency on 1GB VPS for paying subscribers.
• Implemented SQLite WAL concurrency (30s busy-timeout), automated billing pipelines, and hardened Linux VPS deployments.

KEY ENGINEERING PROJECTS

BulkBeat TV — Real-Time Market Intelligence Engine (bulkbeattv.com) Systems Architect & Backend Lead
• Engineered an event-driven ingestion engine in Python AsyncIO, scanning 5+ live exchange/media feeds with sub-5s Telegram Webhook alert delivery.
• Implemented 22-rule deterministic AI scoring eliminating 90%+ market noise; built cookie warming and identity rotation routines to bypass bot blocks.
• Optimized SQLite with Write-Ahead Logging (WAL) and 30s busy-timeout queues; scaled to 6,000+ active users and ₹1.11L revenue within months.

SmartGalla — Geospatial Hyperlocal Commerce Platform Backend & Database Engineer
• Engineered Point-in-Polygon geofencing with in-memory LRU cache, delivering sub-100ms merchant spatial queries and atomic PostgreSQL ledgers.
• Designed multi-tenant schema with 50+ migrations, Row-Level Security (RLS) policies, and atomic credit/cash-on-delivery payment ledgers.
• Piloted across 4–5 local Kirana shops in Kanpur; validated spatial delta replication before sunseting due to hosting infrastructure costs.

College Student Management System (CSMS) — Cloud Educational ERP Backend Engineer
• Developed cloud-native Educational ERP REST API in FastAPI (Python) and PostgreSQL featuring 28 endpoints with automated OpenAPI docs.
• Architected 4-tier Role-Based Access Control (HOD, Section Incharge, Faculty, Student) using JWT authentication and bcrypt password hashing.
• Enforced composite relational constraints (student_id, record_date) for attendance logging (<75% warning trigger) and automated grade evaluation.

DocRoute — Adaptive Document Intelligence Engine Systems Engineer · Status: Under Development
• Architecting modular document extraction engine combining PyMuPDF native vector text parsing with an OCR fallback pipeline for scanned pages.
• Designing quality-assessment routing heuristic that inspects character density and rendering resolution before invoking OCR workers.
• Structuring unified JSON schema with extraction provenance for downstream NLP pipelines (benchmarks reserved until full evaluation).

EDUCATION

Bachelor of Computer Applications (BCA) — Dr. Virendra Swarup Institute of Computer Studies (VSICS), CSJM University, Kanpur
2024 – 2027 · CGPA: 8.0 / 10 · Target SGPA: >= 9.0

CERTIFICATIONS & HONORS

Generative AI & Data Analytics — Simplilearn (Distinction) · ML Internship — CodSoft (Distinction) · Azure Cloud — VSICS · National Hackathon Finalist — ECLearnix (Distinction)